

Direction 1: Audited Primal–Dual Policy Learning

Simultaneous Error Control and Cost Regret

Draft research direction for Cost-Aware PAC Labeling

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Abstract

The current audited online PAC-labeling result controls labeling error but does not compare expert expenditure with a cost-optimal policy. This note develops a candidate conference-paper direction that adds such a comparison. The main change is to maintain a distribution over a fixed finite class of threshold policies in addition to the dual price of error. Expert deferrals and random audits provide inverse-probability estimates of every policy’s counterfactual AI loss. A primal exponential-weights update learns inexpensive policies, while a projected dual update enforces the error budget. Under an unknown i.i.d. data distribution, a fixed policy class, predictable decisions, independent audits, bounded loss, and a safe expert policy, the procedure has high-probability error slack and audit-inclusive cost regret of order $\tilde{O}(T^{2/3})$. A tightened internal error target gives exact control at the requested level ε . The result is distribution-free over the class of i.i.d. distributions, but it is not an arbitrary-sequence guarantee.

1 Problem and policy comparator

We first specialize to one cheap AI source, matching the real online experiment in the current project. At round t , the learner observes non-expert information

$$Z_t = (X_t, \hat{Y}_t, s_t),$$

where $\hat{Y}_t$ is the AI label and $s_t \in [0, 1]$ is a fixed proxy loss or uncertainty score. The expert label Y_t is initially hidden. The bounded AI loss is

$$\ell_t = \ell(Y_t, \hat{Y}_t) \in [0, 1].$$

An expert label is correct and costs one; an unaudited AI label has deployment cost zero. An audited AI label is corrected to Y_t and therefore also costs one expert query.

Fix $K \geq 2$ and thresholds $\tau_1, \dots, \tau_{K-1}$ before the online run. Policy k defers to the expert when $s_t \geq \tau_k$. Include an always-expert policy as policy K . Write

$$D_t(k) = \begin{cases} \mathbf{1}\{s_t \geq \tau_k\}, & k < K, \\ 1, & k = K, \end{cases}$$

and define the counterfactual loss and base deployment cost of policy k by

$$h_t(k) = (1 - D_t(k))\ell_t, \quad c_t(k) = D_t(k). \tag{1}$$

Thus $h_t(k)$ is the error that policy k would incur before any audit correction, and $c_t(k)$ is its ordinary expert-deferral cost. The vectors $\mathbf{h}_t = (h_t(1), \dots, h_t(K))$ and $\mathbf{c}_t = (c_t(1), \dots, c_t(K))$ lie in $[0, 1]^K$.

For $w \in \Delta_K := \{w \in \mathbb{R}_+^K : \sum_k w(k) = 1\}$, let

$$H(w) = \mathbb{E}[w^\top \mathbf{h}_t], \quad C(w) = \mathbb{E}[w^\top \mathbf{c}_t]. \quad (2)$$

The expectation is under the unknown data distribution and the independent randomization $k \sim w$. The population comparator is the randomized-policy linear program

$$C^\star(\varepsilon) = \min_{w \in \Delta_K} C(w) \quad \text{subject to} \quad H(w) \leq \varepsilon. \quad (3)$$

The comparator is fixed and non-clairvoyant: it may know the population but it never observes Y_t before deciding whether to use AI or the expert. Randomized mixtures are included because they convexify the cost–error frontier and permit a small tightening of the error target without a discontinuous jump in optimal cost.

2 Assumptions

The following are a clean sufficient set of assumptions for the first simultaneous error-and-cost theorem. They are deliberately stronger than the assumptions of the current arbitrary-sequence error-only theorem.

Assumption 1 (Unknown i.i.d. stream). *The pairs (Z_t, Y_t) are independent and identically distributed according to an arbitrary unknown distribution P . The AI predictor, proxy score, and threshold grid are fixed before the stream begins.*

Assumption 2 (Bounded loss and a safe action). *The loss satisfies $0 \leq \ell_t \leq 1$. The always-expert policy K has $h_t(K) = 0$ and $c_t(K) = 1$ pointwise. The requested target satisfies $\varepsilon \in (0, 1)$.*

Assumption 3 (Predictability and audit randomization). *At the start of round t , the policy distribution w_t , dual variable λ_t , and audit probability are measurable with respect to the past. The learner samples its policy and chooses AI versus expert using only the past and current Z_t . If AI is selected, the audit draw is conditionally independent Bernoulli with its declared probability and is drawn before observing Y_t .*

Assumption 4 (Counterfactual observability). *Whenever an expert label is obtained, the learner can evaluate $h_t(k)$ for every $k \in [K]$. In the one-source threshold class this is automatic: each policy either uses the same AI label, whose loss is then known, or defers and has loss zero.*

Remark 1 (What is not assumed). *No calibration relation such as $s_t = \mathbb{P}(\hat{Y}_t \neq Y_t \mid Z_t)$ is required. A poor score may make the best threshold in the class expensive, but validity and regret relative to that class remain meaningful. The i.i.d. assumption does rule out unrestricted distribution shift and adversarial data sequences; see Section 7.*

3 Audited primal–dual policy learner

Let $\bar{\varepsilon} \in (0, \varepsilon]$ be an internal error target and let $\Lambda > 0$ be a dual cap. For the fixed-horizon theorem, take a constant audit probability

$$\gamma = T^{-1/3}. \quad (4)$$

The probability is constant within a run but vanishes with the horizon. An anytime implementation can use a doubling schedule or $p_t \asymp t^{-1/3}$ with adaptive learning rates.

When policy J_t is selected, define its actual deferral decision $D_t = D_t(J_t)$. If $D_t = 0$, draw $A_t \sim \text{Bernoulli}(\gamma)$; if $D_t = 1$, set $A_t = 0$ because the expert is already queried. The expert-label observation indicator and its conditional probability are

$$O_t = D_t + (1 - D_t)A_t, \quad Q_t = D_t + (1 - D_t)\gamma. \quad (5)$$

Thus $Q_t \in [\gamma, 1]$. The inverse-probability estimate of every policy's loss is

$$\hat{h}_t(k) = \frac{O_t}{Q_t} h_t(k), \quad k \in [K]. \quad (6)$$

When $O_t = 0$, the implementation sets the entire vector to zero without evaluating the hidden loss.

The primal update uses the estimated Lagrangian loss

$$g_t(k) = c_t(k) + \lambda_t \hat{h}_t(k). \quad (7)$$

The common term $-\lambda_t \bar{\varepsilon}$ is omitted because it does not affect the exponential-weights normalization. The dual update uses

$$\hat{v}_t = w_t^\top \hat{\mathbf{h}}_t - \bar{\varepsilon}. \quad (8)$$

Algorithm 1 Audited primal–dual policy learning

Require: Horizon T ; threshold policies $[K]$ including always-expert; target $\bar{\varepsilon}$; audit rate γ ; dual cap Λ ; learning rates η_w, η_λ .

- 1: Initialize $w_1(k) = 1/K$ and $\lambda_1 = 0$.
- 2: **for** $t = 1, \dots, T$ **do**
- 3: Observe $Z_t = (X_t, \hat{Y}_t, s_t)$ and compute $\{D_t(k), c_t(k)\}_{k=1}^K$.
- 4: Sample $J_t \sim w_t$ and set $D_t = D_t(J_t)$.
- 5: **if** $D_t = 1$ **then**
- 6: Query the expert, release $\tilde{Y}_t = Y_t$, and set $A_t = 0$, $O_t = Q_t = 1$.
- 7: **else**
- 8: Draw $A_t \sim \text{Bernoulli}(\gamma)$.
- 9: **if** $A_t = 1$ **then**
- 10: Query the expert, release the corrected label $\tilde{Y}_t = Y_t$, and set $O_t = 1$, $Q_t = \gamma$.
- 11: **else**
- 12: Release $\tilde{Y}_t = \hat{Y}_t$ and set $O_t = 0$, $Q_t = \gamma$.
- 13: **end if**
- 14: **end if**
- 15: Form $\hat{\mathbf{h}}_t$ by (6).
- 16: Set $g_t(k) = c_t(k) + \lambda_t \hat{h}_t(k)$ and update

$$w_{t+1}(k) = \frac{w_t(k) e^{-\eta_w g_t(k)}}{\sum_{j=1}^K w_t(j) e^{-\eta_w g_t(j)}}.$$

- 17: Set $\hat{v}_t = w_t^\top \hat{\mathbf{h}}_t - \bar{\varepsilon}$ and update

$$\lambda_{t+1} = \Pi_{[0, \Lambda]} (\lambda_t + \eta_\lambda \hat{v}_t).$$

- 18: **end for**
-

The counterfactual and final released-label losses of the selected policy are

$$H_t = h_t(J_t), \quad G_t = \ell(Y_t, \tilde{Y}_t). \quad (9)$$

Pointwise, $G_t \leq H_t$: expert deferrals have zero loss, and audited AI mistakes are corrected. The realized total expert cost is

$$C_t^{\text{tot}} = D_t + (1 - D_t)A_t. \quad (10)$$

This counts an audited-and-corrected AI item once, not twice.

4 Explicit finite-horizon quantities

The theorem below gives explicit, intentionally conservative constants. Fix $\delta \in (0, 1)$ and define

$$x = \log\left(\frac{64K}{\delta}\right). \quad (11)$$

For $\gamma \in (0, 1]$, let

$$V_T := \frac{T}{\gamma} + \sqrt{\frac{2Tx}{\gamma^3}} + \frac{x}{3\gamma^2}, \quad (12)$$

$$S_w := 2T + 2\Lambda^2 V_T, \quad S_\lambda := 2T + 2V_T. \quad (13)$$

Choose

$$\eta_w := \min\left\{\frac{1}{2(1 + \Lambda/\gamma)}, \sqrt{\frac{2 \log K}{S_w}}\right\}, \quad (14)$$

$$\eta_\lambda := \frac{\Lambda}{\sqrt{S_\lambda}}. \quad (15)$$

Define the primal and dual optimization terms

$$R_w := \frac{\log K}{\eta_w} + \frac{\eta_w}{2} S_w, \quad (16)$$

$$R_\lambda := \frac{\Lambda^2}{2\eta_\lambda} + \frac{\eta_\lambda}{2} S_\lambda = \Lambda \sqrt{S_\lambda}, \quad (17)$$

and the stochastic conversion term

$$R_{\text{stoch}} := 2\sqrt{2Tx} + 2\Lambda \left(\sqrt{\frac{2Tx}{\gamma}} + \frac{x}{3\gamma} \right). \quad (18)$$

Finally, set

$$e_T := \frac{R_w + R_\lambda + R_{\text{stoch}}}{T}, \quad r_T := \sqrt{\frac{x}{2T}}, \quad a_T := \gamma + \sqrt{\frac{2\gamma x}{T}} + \frac{x}{3T}. \quad (19)$$

Here e_T is a population saddle-point error, r_T converts population loss and cost to the realized stream, and a_T upper-bounds the fraction of extra AI audits.

5 Main bounds

For a working target $b \in (0, 1)$, define the population value

$$C^*(b) = \min_{w \in \Delta_K} \{C(w) : H(w) \leq b\}.$$

Theorem 1 (Simultaneous finite-horizon bounds). *Suppose Assumptions 1–4 hold. Run Algorithm 1 with internal target $\bar{\varepsilon} \in (0, 1)$, $\gamma \in (0, 1]$, and $\Lambda > 1/\bar{\varepsilon}$, using (14)–(15). Then, with probability at least $1 - \delta$ over the data, policy randomization, and audits,*

$$\frac{1}{T} \sum_{t=1}^T H_t \leq \bar{\varepsilon} + \frac{e_T}{\Lambda - 1/\bar{\varepsilon}} + r_T, \quad (20)$$

$$\frac{1}{T} \sum_{t=1}^T G_t \leq \frac{1}{T} \sum_{t=1}^T H_t, \quad (21)$$

$$\frac{1}{T} \sum_{t=1}^T C_t^{\text{tot}} \leq C^*(\bar{\varepsilon}) + e_T + r_T + a_T. \quad (22)$$

Thus the result controls the more demanding counterfactual pre-correction loss, and its cost comparison includes all expert deferrals and audits.

The base theorem has a vanishing error slack. The next corollary spends this slack in advance to achieve the requested error level exactly.

Corollary 1 (Exact control at ε and regret to $C^*(\varepsilon)$). *Set*

$$\Lambda = \frac{4}{\varepsilon}, \quad \kappa_T = \frac{\varepsilon e_T}{2} + r_T, \quad \bar{\varepsilon} = \varepsilon - \kappa_T. \quad (23)$$

Assume $\kappa_T \leq \varepsilon/2$. Then, on the event of Theorem 1,

$$\frac{1}{T} \sum_{t=1}^T G_t \leq \frac{1}{T} \sum_{t=1}^T H_t \leq \varepsilon, \quad (24)$$

$$\frac{1}{T} \sum_{t=1}^T C_t^{\text{tot}} \leq C^*(\varepsilon) + e_T + \frac{\kappa_T}{\varepsilon} + r_T + a_T. \quad (25)$$

In particular, for $\gamma = T^{-1/3}$ and fixed (ε, K, δ) ,

$$\sum_{t=1}^T H_t \leq \varepsilon T, \quad (26)$$

$$\sum_{t=1}^T C_t^{\text{tot}} \leq T C^*(\varepsilon) + \tilde{O}_\varepsilon(T^{2/3}). \quad (27)$$

The notation $\tilde{O}_\varepsilon$ hides logarithmic factors in $K, T, 1/\delta$ and constants polynomial in $1/\varepsilon$.

Remark 2 (Why $T^{-1/3}$). The extra audit cost is approximately $T\gamma$, while the partial-feedback learning term is approximately $\sqrt{T/\gamma}$. Balancing these terms gives $\gamma \asymp T^{-1/3}$ and cumulative order $T^{2/3}$. A fixed positive audit rate would instead incur linear audit cost.

6 Proofs

We give all intermediate steps because the distinction between observed, counterfactual, and released-label loss is central to the result.

6.1 Unbiased loss estimates and second moments

Let $\mathcal{F}_{t-1}$ contain the complete history through round $t-1$. For analysis, let $\mathcal{G}_t$ augment $\mathcal{F}_{t-1}$ with the current data, selected policy, all current counterfactual losses, and the observation probability Q_t , but not the audit coin when AI is selected. Conditional on $\mathcal{G}_t$, O_t is Bernoulli with mean Q_t ; on a deferral, this is the degenerate Bernoulli variable equal to one.

Lemma 1 (Inverse-probability identities). *For every $k \in [K]$,*

$$\mathbb{E}[\widehat{h}_t(k) \mid \mathcal{G}_t] = h_t(k), \quad (28)$$

$$\mathbb{E}[\widehat{h}_t(k)^2 \mid \mathcal{G}_t] \leq \frac{h_t(k)^2}{\gamma}. \quad (29)$$

Moreover,

$$\mathbb{E}[w_t^\top \widehat{\mathbf{h}}_t \mid \mathcal{F}_{t-1}] = H(w_t), \quad (30)$$

$$\mathbb{E}[w_t^\top (\widehat{\mathbf{h}}_t \odot \widehat{\mathbf{h}}_t) \mid \mathcal{F}_{t-1}] \leq \frac{1}{\gamma}. \quad (31)$$

Here $\odot$ denotes componentwise multiplication.

Proof. Conditional on $\mathcal{G}_t$, $h_t(k)$ and Q_t are fixed and $\mathbb{E}[O_t \mid \mathcal{G}_t] = Q_t$. Hence

$$\mathbb{E}\left[\frac{O_t}{Q_t} h_t(k) \mid \mathcal{G}_t\right] = h_t(k).$$

Because $O_t^2 = O_t$ and $Q_t \geq \gamma$,

$$\mathbb{E}\left[\left(\frac{O_t}{Q_t} h_t(k)\right)^2 \mid \mathcal{G}_t\right] = \frac{h_t(k)^2}{Q_t} \leq \frac{h_t(k)^2}{\gamma}.$$

Taking iterated expectations over the current i.i.d. data and selected policy gives (30). Summing (29) with weights $w_t(k)$ and using $0 \leq h_t(k) \leq 1$ gives (31). $\square$

The following variance event is what improves the rate over a crude $1/\gamma$ -range analysis.

Lemma 2 (High-probability quadratic variation). *With probability at least $1 - e^{-x}$,*

$$\sum_{t=1}^T w_t^\top (\widehat{\mathbf{h}}_t \odot \widehat{\mathbf{h}}_t) \leq V_T, \quad (32)$$

where V_T is defined in (12). On this event,

$$\sum_{t=1}^T w_t^\top (\mathbf{g}_t \odot \mathbf{g}_t) \leq S_w, \quad (33)$$

$$\sum_{t=1}^T \widehat{v}_t^2 \leq S_\lambda. \quad (34)$$

Proof. Set $X_t = w_t^\top (\hat{\mathbf{h}}_t \odot \hat{\mathbf{h}}_t)$. By Lemma 1, $\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] \leq 1/\gamma$. Also $0 \leq X_t \leq 1/\gamma^2$. Conditional on $\mathcal{G}_t$, $X_t = (O_t/Q_t^2)w_t^\top (\mathbf{h}_t \odot \mathbf{h}_t)$, so

$$\mathbb{E}[X_t^2 \mid \mathcal{G}_t] \leq \frac{1}{Q_t^3} \leq \frac{1}{\gamma^3}.$$

Therefore the martingale differences $X_t - \mathbb{E}[X_t \mid \mathcal{F}_{t-1}]$ have increments bounded above by $1/\gamma^2$ and total conditional variance at most T/γ^3 . Freedman's inequality gives

$$\sum_{t=1}^T X_t \leq \frac{T}{\gamma} + \sqrt{\frac{2Tx}{\gamma^3}} + \frac{x}{3\gamma^2} = V_T.$$

Next, $(a+b)^2 \leq 2a^2 + 2b^2$ gives

$$g_t(k)^2 \leq 2c_t(k)^2 + 2\Lambda^2 \hat{\mathbf{h}}_t(k)^2.$$

Summing with weights $w_t(k)$ proves (33). Finally, Jensen's inequality gives $(w_t^\top \hat{\mathbf{h}}_t)^2 \leq w_t^\top (\hat{\mathbf{h}}_t \odot \hat{\mathbf{h}}_t)$, and hence

$$\hat{v}_t^2 \leq 2(w_t^\top \hat{\mathbf{h}}_t)^2 + 2\bar{\varepsilon}^2.$$

Summation proves (34). □

6.2 Primal and dual regret

Define the estimated per-round Lagrangian

$$\hat{F}_t(w, \lambda) = w^\top \mathbf{c}_t + \lambda(w^\top \hat{\mathbf{h}}_t - \bar{\varepsilon}). \quad (35)$$

Lemma 3 (Estimated saddle regret). *On the event in Lemma 2, for every $u \in \Delta_K$ and $\lambda \in [0, \Lambda]$,*

$$\sum_{t=1}^T \hat{F}_t(w_t, \lambda) - \sum_{t=1}^T \hat{F}_t(u, \lambda_t) \leq R_w + R_\lambda. \quad (36)$$

Proof. Because $g_t(k) \geq 0$ and $g_t(k) \leq 1 + \Lambda/\gamma$, the first term in the minimum defining η_w ensures $\eta_w g_t(k) \leq 1/2$. The standard exponential-weights potential argument, using $e^{-z} \leq 1 - z + z^2/2$ for $z \geq 0$, gives

$$\sum_{t=1}^T w_t^\top \mathbf{g}_t - \sum_{t=1}^T u^\top \mathbf{g}_t \leq \frac{\log K}{\eta_w} + \frac{\eta_w}{2} \sum_{t=1}^T w_t^\top (\mathbf{g}_t \odot \mathbf{g}_t) \leq R_w.$$

Since the term $-\lambda_t \bar{\varepsilon}$ is common to w_t and u , this is equivalently

$$\sum_{t=1}^T \hat{F}_t(w_t, \lambda_t) - \sum_{t=1}^T \hat{F}_t(u, \lambda_t) \leq R_w.$$

For the projected dual-ascent step, nonexpansiveness of Euclidean projection implies, for every $\lambda \in [0, \Lambda]$,

$$\sum_{t=1}^T (\lambda - \lambda_t) \hat{v}_t \leq \frac{\Lambda^2}{2\eta_\lambda} + \frac{\eta_\lambda}{2} \sum_{t=1}^T \hat{v}_t^2 \leq R_\lambda.$$

The left side equals $\sum_t \hat{F}_t(w_t, \lambda) - \sum_t \hat{F}_t(w_t, \lambda_t)$. Adding the primal and dual inequalities proves (36). □

6.3 From sampled Lagrangians to the population saddle problem

Define the population Lagrangian

$$F(w, \lambda) = C(w) + \lambda(H(w) - \bar{\varepsilon}). \quad (37)$$

Lemma 4 (Uniform stochastic conversion). *With probability at least $1 - 8Ke^{-x}$, simultaneously for every $u \in \Delta_K$ and $\lambda \in [0, \Lambda]$,*

$$\left| \sum_{t=1}^T \{ \widehat{F}_t(w_t, \lambda) - F(w_t, \lambda) \} \right| \leq \sqrt{2Tx} + \Lambda \left(\sqrt{\frac{2Tx}{\gamma}} + \frac{x}{3\gamma} \right), \quad (38)$$

$$\left| \sum_{t=1}^T \{ \widehat{F}_t(u, \lambda_t) - F(u, \lambda_t) \} \right| \leq \sqrt{2Tx} + \Lambda \left(\sqrt{\frac{2Tx}{\gamma}} + \frac{x}{3\gamma} \right). \quad (39)$$

Proof. We separate cost and loss. Because w_t is measurable with respect to the past and the data are i.i.d., $w_t^\top \mathbf{c}_t - C(w_t)$ is a bounded martingale difference. Azuma–Hoeffding controls its absolute sum by $\sqrt{2Tx}$ except on an event of probability at most $2e^{-x}$. For any fixed policy k , the same argument controls $\sum_t (c_t(k) - C(e_k))$. A union bound over k and linearity in u extend the result to every $u \in \Delta_K$.

For loss, Lemma 1 shows that $w_t^\top \widehat{\mathbf{h}}_t - H(w_t)$ is a martingale difference. Its increment is bounded in absolute value by $1/\gamma$, and its conditional variance is at most $1/\gamma$. Freedman’s inequality therefore gives

$$\left| \sum_{t=1}^T \{ w_t^\top \widehat{\mathbf{h}}_t - H(w_t) \} \right| \leq \sqrt{\frac{2Tx}{\gamma}} + \frac{x}{3\gamma}$$

outside an event of probability at most $2e^{-x}$. Multiplying by any fixed $\lambda \in [0, \Lambda]$ costs at most a factor Λ .

For the right side, fix a vertex e_k of the simplex. The sequence

$$\lambda_t \{ \widehat{h}_t(k) - H(e_k) \}$$

is a martingale difference with absolute increment at most Λ/γ and conditional variance at most Λ^2/γ . Freedman’s inequality gives the stated loss bound. A union bound over k and linearity in u give the result for every mixture u . Combining the cost and loss events proves the claim; the displayed failure probability is a conservative union bound. $\square$

Proposition 1 (Population saddle gap). *Let*

$$\bar{w}_T = \frac{1}{T} \sum_{t=1}^T w_t, \quad \bar{\lambda}_T = \frac{1}{T} \sum_{t=1}^T \lambda_t.$$

With probability at least $1 - \delta/2$,

$$\max_{\lambda \in [0, \Lambda]} F(\bar{w}_T, \lambda) - \min_{u \in \Delta_K} F(u, \bar{\lambda}_T) \leq e_T. \quad (40)$$

Proof. Intersect the events of Lemmas 2 and 4. Their total failure probability is below $\delta/2$ under the choice (11). Combining Lemma 3 with the two conversion bounds gives, for every u and λ ,

$$\sum_{t=1}^T F(w_t, \lambda) - \sum_{t=1}^T F(u, \lambda_t) \leq R_w + R_\lambda + R_{\text{stoch}}.$$

The population Lagrangian is bilinear, so the left side divided by T equals $F(\bar{w}_T, \lambda) - F(u, \bar{\lambda}_T)$. Maximizing over λ and minimizing over u proves the proposition. $\square$

6.4 Exact penalty and separate population bounds

Lemma 5 (Safe policy bounds the optimal dual price). *For any target $b \in (0, 1)$, the LP defining $C^*(b)$ has an optimal dual price λ_b^* satisfying*

$$0 \leq \lambda_b^* \leq \frac{1}{b}. \quad (41)$$

Proof. The problem is a finite linear program and is strictly feasible because the always-expert policy has loss zero. Strong duality therefore holds. Its dual function is

$$d_b(\lambda) = \min_{w \in \Delta_K} \{C(w) + \lambda(H(w) - b)\}.$$

Evaluating the inner minimum at the always-expert policy gives $d_b(\lambda) \leq 1 - \lambda b$. At a dual optimum, $C^*(b) = d_b(\lambda_b^*) \geq 0$, so $0 \leq 1 - \lambda_b^* b$, proving $\lambda_b^* \leq 1/b$. $\square$

Lemma 6 (Saddle gap implies cost and feasibility). *Suppose (40) holds for target b and $\Lambda > \lambda_b^*$. Then*

$$C(\bar{w}_T) \leq C^*(b) + e_T, \quad (42)$$

$$H(\bar{w}_T) \leq b + \frac{e_T}{\Lambda - \lambda_b^*}. \quad (43)$$

Proof. Let w_b^* be a primal optimum. Since it is feasible and $\bar{\lambda}_T \geq 0$,

$$\min_u F(u, \bar{\lambda}_T) \leq F(w_b^*, \bar{\lambda}_T) \leq C^*(b).$$

Therefore (40) implies

$$C(\bar{w}_T) + \Lambda[H(\bar{w}_T) - b]_+ \leq C^*(b) + e_T. \quad (44)$$

Setting the dual price to zero inside the maximum gives (42).

Strong duality at λ_b^* gives, for every w ,

$$C(w) + \lambda_b^*(H(w) - b) \geq C^*(b).$$

If $H(\bar{w}_T) \leq b$, there is nothing to prove. Otherwise, put $v = H(\bar{w}_T) - b > 0$. Then

$$C(\bar{w}_T) + \Lambda v \geq C^*(b) + (\Lambda - \lambda_b^*)v.$$

Combining this inequality with (44) yields $v \leq e_T/(\Lambda - \lambda_b^*)$. $\square$

6.5 Realized loss, realized cost, and proof of the main theorem

Lemma 7 (Online-to-realized conversion). *With probability at least $1 - 6e^{-x}$,*

$$\frac{1}{T} \sum_{t=1}^T H_t \leq H(\bar{w}_T) + r_T, \quad (45)$$

$$\frac{1}{T} \sum_{t=1}^T c_t(J_t) \leq C(\bar{w}_T) + r_T, \quad (46)$$

$$\frac{1}{T} \sum_{t=1}^T (1 - D_t) A_t \leq a_T. \quad (47)$$

Proof. Conditional on the past, the current data are a fresh i.i.d. draw and $J_t \sim w_t$. Hence

$$\mathbb{E}[H_t \mid \mathcal{F}_{t-1}] = H(w_t), \quad \mathbb{E}[c_t(J_t) \mid \mathcal{F}_{t-1}] = C(w_t).$$

Both realized variables lie in $[0, 1]$. Hoeffding–Azuma gives (45) and (46) with the stated r_T after a union bound.

For $B_t = (1 - D_t)A_t$, predictability and conditional audit independence give

$$\mathbb{E}[B_t \mid \text{pre-audit information}] = (1 - D_t)\gamma \leq \gamma.$$

The increments are Bernoulli and have conditional variance at most γ . Freedman’s inequality yields

$$\sum_{t=1}^T B_t \leq \gamma T + \sqrt{2\gamma T x} + \frac{x}{3},$$

which is (47). □

Proof of Theorem 1. On the population saddle event, Lemmas 5 and 6, with $b = \bar{\varepsilon}$, give

$$C(\bar{w}_T) \leq C^\star(\bar{\varepsilon}) + e_T, \quad H(\bar{w}_T) \leq \bar{\varepsilon} + \frac{e_T}{\Lambda - 1/\bar{\varepsilon}}.$$

The latter uses $\lambda_{\bar{\varepsilon}}^\star \leq 1/\bar{\varepsilon}$. Intersect this event with Lemma 7; the definition of x makes the combined failure probability at most δ . Equation (45) proves (20). Audit-and-correct gives $G_t \leq H_t$ pointwise, proving (21). Finally,

$$C_t^{\text{tot}} = c_t(J_t) + (1 - D_t)A_t.$$

Combining (46), (47), and the population cost bound proves (22). □

6.6 Tightening and proof of the exact-error corollary

Lemma 8 (Cost of tightening). *For every $0 \leq \kappa \leq \varepsilon$,*

$$C^\star(\varepsilon - \kappa) \leq C^\star(\varepsilon) + \frac{\kappa}{\varepsilon}. \quad (48)$$

Proof. Let $w_\varepsilon^\star$ solve (3), let e_K be the always-expert policy, and set $\alpha = \kappa/\varepsilon$. The mixture

$$w' = (1 - \alpha)w_\varepsilon^\star + \alpha e_K$$

satisfies

$$H(w') = (1 - \alpha)H(w_\varepsilon^\star) \leq (1 - \alpha)\varepsilon = \varepsilon - \kappa.$$

Its cost obeys

$$C(w') = (1 - \alpha)C^\star(\varepsilon) + \alpha C(e_K) \leq C^\star(\varepsilon) + \alpha,$$

because $C(e_K) = 1$. Optimality at level $\varepsilon - \kappa$ proves the claim. $\square$

Proof of Corollary 1. Since $\kappa_T \leq \varepsilon/2$, we have $\bar{\varepsilon} \geq \varepsilon/2$. Consequently

$$\Lambda - \frac{1}{\bar{\varepsilon}} \geq \frac{4}{\varepsilon} - \frac{2}{\varepsilon} = \frac{2}{\varepsilon}.$$

Theorem 1 therefore gives

$$\frac{1}{T} \sum_{t=1}^T H_t \leq \bar{\varepsilon} + \frac{\varepsilon e_T}{2} + r_T = \varepsilon,$$

which proves (24). Lemma 8 and (22) give

$$\frac{1}{T} \sum_{t=1}^T C_t^{\text{tot}} \leq C^\star(\varepsilon) + \frac{\kappa_T}{\varepsilon} + e_T + r_T + a_T.$$

It remains to check the rate. With $\gamma = T^{-1/3}$, $V_T = \tilde{O}(T^{4/3})$. Hence $S_w = \tilde{O}(\Lambda^2 T^{4/3})$ and $S_\lambda = \tilde{O}(T^{4/3})$. The learning rates in (14)–(15) give

$$R_w + R_\lambda + R_{\text{stoch}} = \tilde{O}((1 + \Lambda)T^{2/3}).$$

Thus $e_T = \tilde{O}((1 + \Lambda)T^{-1/3})$ and $a_T = O(T^{-1/3})$ up to logarithmic terms. Substituting $\Lambda = 4/\varepsilon$ and multiplying the average-cost bound by T proves (27). $\square$

7 Interpretation and scope

What the theorem adds. The existing online theorem controls error for any data sequence but does not compare cost with an oracle. The present direction adds a primal policy learner and proves a comparison with the cheapest fixed randomized threshold policy whose population counterfactual loss is at most ε . All expert audits are charged to the algorithm.

Meaning of distribution-free. The result is uniform over all i.i.d. distributions P ; it assumes no parametric model and no calibrated proxy. It is not distribution-free in the stronger arbitrary-sequence sense. The i.i.d. assumption is what lets past audits reveal the future population cost and loss of a fixed policy.

Why the comparator uses counterfactual loss. Audited AI mistakes are corrected, so their final released-label loss is zero. Using final loss alone would make an unsafe AI source appear perfect whenever it is audited. The constraint therefore controls H_t , the loss the selected AI action would have incurred before correction. Since $G_t \leq H_t$, this automatically controls the released labels as well.

Fixed scores and policies. The first theorem freezes the AI model, score function, and threshold grid. Allowing the proxy itself to be retrained from past audits is a distinct adaptive-policy problem. It would require either sample splitting, a complexity-controlled adaptive class, or a sequential uniform-convergence argument.

Several cheap sources. The proof extends algebraically to a finite class of multi-source routing policies if an observed expert label reveals the counterfactual loss of every candidate policy. In practice this may require evaluating every cheap model. Those model-call costs must then be included in $c_t(k)$. Without such full counterfactual observability, the feedback becomes a contextual bandit problem and the importance weights and regret rate change.

Conference-paper status. The threshold LP, cost-sensitive routing, and inverse-probability auditing are not individually novelty claims. A defensible contribution would be the simultaneous high-probability error and audit-inclusive cost guarantee, ideally paired with a lower bound showing whether the $T^{2/3}$ gap-free rate is optimal and experiments that report every component of total cost.